CIE International A Level Physics

Topic 22

Quantum Physics

Revision Booklet

This booklet covers:

- The Photoelectric Effect
- Photon Energy and the Planck Relation
- Einstein's Photoelectric Equation
- Wave–Particle Duality & de Broglie Wavelength
- Energy Levels and Line Spectra
- X-Ray Spectra (characteristic and continuous)

Core Concepts and Definitions

The Quantum Model of Light

Classical wave theory cannot explain certain phenomena involving light and matter. Quantum theory proposes that electromagnetic radiation is emitted and absorbed in discrete packets of energy called **photons**.

- A **photon** is a quantum of electromagnetic energy.
- The energy of a photon depends only on its **frequency**, not its intensity.
- Intensity is proportional to the **number of photons** per unit time, not the energy of each photon.

Photon Energy — The Planck Relation

$$E = hf = \frac{hc}{\lambda}$$

E = energy of one photon (J)
h = Planck's constant = 6.63×10^{-34} J s
f = frequency of the radiation (Hz)
c = speed of light = 3.00×10^{8} m s^{-1}
λ = wavelength (m)

The Electronvolt

The **electronvolt** (eV) is a convenient unit of energy at the atomic scale.

$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$$

To convert eV $\rightarrow$ J: multiply by 1.60×10^{-19}.
To convert J $\rightarrow$ eV: divide by 1.60×10^{-19}.

The Photoelectric Effect

The Photoelectric Effect

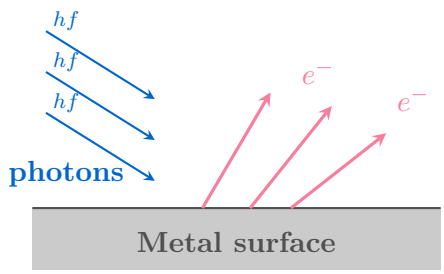
The **photoelectric effect** is the emission of electrons from a metal surface when electromagnetic radiation of sufficiently high frequency is incident on it.

- Electrons emitted are called **photoelectrons**.
- Emission occurs **instantaneously** if the frequency is above a threshold — there is no time delay.
- Below the **threshold frequency** f_0, no electrons are emitted regardless of intensity.
- Above f_0, increasing the intensity increases the **number** of photoelectrons per second, not their energy.

Why Classical Wave Theory Fails

- Wave theory predicts that any frequency of sufficient intensity should eventually eject electrons — **not observed**.
- Wave theory predicts a time delay before emission as energy builds up — **not observed** (emission is instantaneous).
- Wave theory predicts that intensity should increase the electron's kinetic energy — **not observed**.

These failures led Einstein to propose the photon model.



Einstein's Photoelectric Equation

$$hf = \phi + \tfrac{1}{2}mv_{\text{max}}^2$$

hf	= energy of the incident photon (J)
ϕ	= **work function** — minimum energy to remove an electron from the surface (J)
$\frac{1}{2}mv_{\text{max}}^2$	= maximum kinetic energy of emitted photoelectron (J)

The **work function** $\phi = hf_0$, where f_0 is the threshold frequency.
Maximum KE can also be written as eV_s, where V_s is the **stopping potential**.

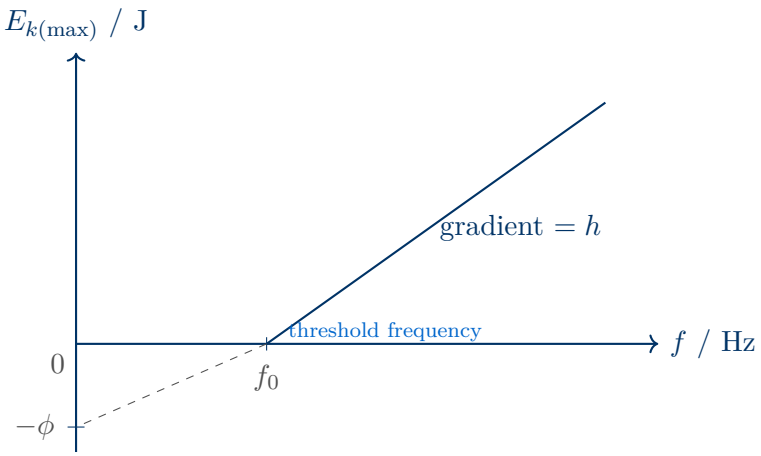
Stopping Potential V_s

The **stopping potential** is the minimum opposing potential difference required to prevent all photoelectrons from reaching the collecting electrode.

$$eV_s = \tfrac{1}{2}mv_{\text{max}}^2 = hf - \phi$$

Measuring V_s at different frequencies allows h and ϕ to be determined experimentally.

Graph of Maximum KE against Frequency



Reading the Graph

- The **gradient** of the $E_{k(\max)}$ vs f graph equals Planck's constant h.
- The **x-intercept** gives the threshold frequency f_0.
- The **y-intercept** (extrapolated) gives $-\phi$ (the negative of the work function).
- Different metals give **parallel lines** — same gradient (h), different intercepts (ϕ).

Wave–Particle Duality

Wave–Particle Duality

Wave–particle duality is the property of matter and radiation by which they exhibit both wave-like and particle-like behaviour depending on the experimental context.

- Light behaves as a **wave**: diffraction, interference, polarisation.
- Light behaves as **particles (photons)**: photoelectric effect, Compton scattering.
- Electrons (and other particles) also show **wave behaviour**: electron diffraction.

de Broglie Wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

λ = de Broglie wavelength (m)
h = Planck's constant (6.63×10^{-34} J s)
p = momentum of the particle (kg m s^{-1})
m = mass of the particle (kg)
v = speed of the particle (m s^{-1})

Electron Diffraction

When electrons are accelerated through a potential difference V and directed at a thin crystal or graphite film, a **diffraction pattern** of rings is observed — confirming their wave nature.

- The kinetic energy gained: $\frac{1}{2}mv^2 = eV$, giving $v = \sqrt{2eV/m}$.
- Therefore: $\lambda = \dfrac{h}{\sqrt{2meV}}$
- Increasing V increases p, decreasing $\lambda \Rightarrow$ rings become **smaller**.
- The wavelength must be comparable to the atomic spacing for diffraction to occur ($\sim 10^{-10}$ m).

Common Mistake — Confusing Photon and Particle Momentum

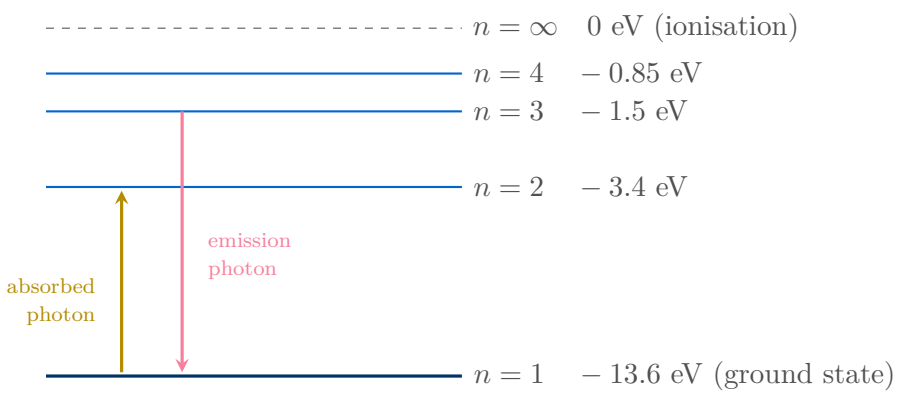
For a **photon** (massless): $p = E/c = hf/c = h/\lambda$.
For a **particle with mass**: $p = mv = h/\lambda$ (de Broglie).
Both use $\lambda = h/p$, but the relationship between energy and momentum differs. Do not use $E = hf$ for material particles.

Energy Levels and Line Spectra

Atomic Energy Levels

Electrons in an atom can only occupy discrete **energy levels**. These are quantised — only specific energies are allowed.

- The **ground state** is the lowest energy level (most stable; most negative value).
- **Excited states** have higher (less negative) energy.
- The **ionisation energy** is the energy required to remove an electron from the ground state to infinity ($E = 0$).
- Energy levels are usually expressed in electronvolts (eV) and are **negative** (bound states).



Hydrogen Energy Level Diagram (schematic)

Photon Energy from a Transition

When an electron moves between energy levels E_1 (lower) and E_2 (higher):

$$hf = E_2 - E_1$$

- **Emission**: electron falls to lower level ⇒ photon **emitted** with $hf = E_2 - E_1$.
- **Absorption**: electron promoted to higher level ⇒ photon **absorbed** with $hf = E_2 - E_1$.

Emission and Absorption Spectra

Line Spectra

- **Emission spectrum**: a series of **bright coloured lines** on a dark background. Each line corresponds to a specific photon frequency emitted during a downward electron transition. Produced by excited gases.
- **Absorption spectrum**: a **continuous spectrum** crossed by dark lines at the same frequencies as the emission lines. Cool gas in front of a broad-spectrum source absorbs specific photon energies.
- The **line positions are unique** to each element — used for identification (spectroscopy).

Why Line Spectra Prove Quantisation

Because electrons can only occupy discrete energy levels, only photons of specific energies (frequencies) can be absorbed or emitted. This produces a **line** spectrum rather than a continuous one. A continuous spectrum would imply electrons can have any energy — which they cannot.

X-Ray Spectra

Production of X-Rays

X-rays are produced when fast-moving electrons are rapidly decelerated by a metal target (anode) in an X-ray tube. Two distinct processes contribute to the X-ray spectrum:

- **Continuous (bremsstrahlung) spectrum**: caused by deceleration of electrons; produces a range of wavelengths with a sharp **minimum wavelength** λ_{min}.
- **Characteristic spectrum**: caused by electron transitions in target atoms; produces sharp **peaks** at specific wavelengths.

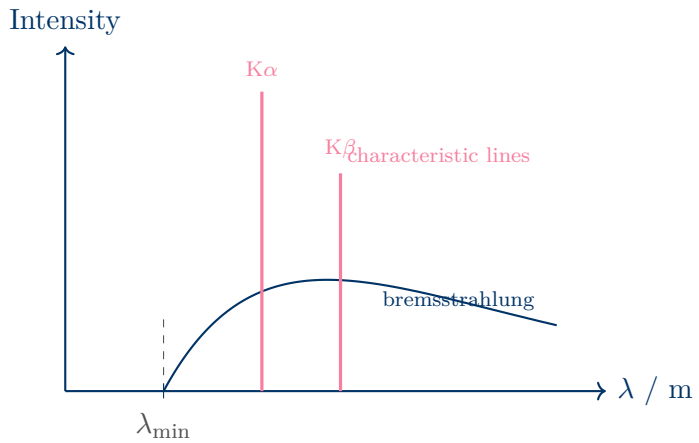
Minimum Wavelength of X-Rays

When an electron gives up all its kinetic energy to a single photon:

$$eV = hf_{\text{max}} = \frac{hc}{\lambda_{\text{min}}}$$

$$\lambda_{\text{min}} = \frac{hc}{eV}$$

where V is the accelerating voltage of the X-ray tube.
Increasing V decreases λ_{min} and shifts the spectrum to shorter wavelengths.



Characteristic X-Ray Lines

- An incident electron knocks out an inner-shell electron, creating a vacancy.
- An outer-shell electron falls into the vacancy, emitting a photon with energy equal to the difference in shell energies.
- Because energy levels are fixed for a given element, the emitted photon has a **fixed, characteristic wavelength**.
- Characteristic lines are **element-specific**: changing the target metal changes the line positions, not λ_{min}.

Formula Summary Sheet

Formula	Quantity	Units
$E = hf$	Photon energy	J
$E = hc/\lambda$	Photon energy from wavelength	J
$hf = \phi + \frac{1}{2}mv_{\text{max}}^2$	Einstein's photoelectric equation	J
$\phi = hf_0$	Work function	J
$eV_s = \frac{1}{2}mv_{\text{max}}^2$	Stopping potential	J, V
$\lambda = h/p = h/mv$	de Broglie wavelength	m
$\lambda = h/\sqrt{2meV}$	de Broglie λ from accelerating pd	m
$hf = E_2 - E_1$	Photon from energy transition	J
$\lambda_{\text{min}} = hc/eV$	Minimum X-ray wavelength	m

Constants: $h = 6.63 \times 10^{-34}$ J s, $c = 3.00 \times 10^8$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg

Exam Technique and Problem-Solving Strategy

Step-by-Step Strategy for Calculation Questions

1. **Identify** the relevant formula — is it a photon, a particle, or a transition?
2. **Convert units**: eV to J ($\times 1.60 \times 10^{-19}$); nm to m ($\times 10^{-9}$).
3. **Substitute** values carefully, showing all working.
4. **Quote** the final answer in appropriate units with correct significant figures.

Common Errors — Avoid These!

- Using the **wrong energy unit** — forgetting to convert eV to J before substituting.
- Confusing **threshold frequency** f_0 with threshold wavelength: a higher f_0 corresponds to a **shorter** threshold wavelength λ_0.
- Applying **intensity** changes to change photon energy — intensity only changes the **number** of photons.
- Using $E = hf$ for the de Broglie relation — this is for photons only; for matter use

$\lambda = h/mv$.

- Forgetting that energy levels are **negative** — the transition energy is $|E_2 - E_1|$, not just $E_2 - E_1$ as magnitudes.
- Confusing **emission** (bright lines) and **absorption** (dark lines) spectra.

Worked Examples

Example 1 — Photoelectric Effect Calculation

Question: Light of wavelength 250 nm is incident on a metal surface with a work function of 4.5 eV. Calculate the maximum kinetic energy of emitted photoelectrons and the stopping potential.

Solution

Solution:
Convert work function: $\phi = 4.5 \times 1.60 \times 10^{-19} = 7.20 \times 10^{-19}$ J
Photon energy:

$$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{250 \times 10^{-9}} = 7.96 \times 10^{-19} \text{ J}$$

Maximum KE:

$$E_{k(\max)} = hf - \phi = 7.96 \times 10^{-19} - 7.20 \times 10^{-19} = \mathbf{7.6 \times 10^{-20}} \text{ J}$$

Stopping potential:

$$V_s = \frac{E_{k(\max)}}{e} = \frac{7.6 \times 10^{-20}}{1.60 \times 10^{-19}} = \mathbf{0.48} \text{ V}$$

Example 2 — de Broglie Wavelength

Question: An electron is accelerated from rest through a potential difference of 3.0 kV. Calculate its de Broglie wavelength.

Solution

Solution:
Energy gained: $E_k = eV = 1.60 \times 10^{-19} \times 3000 = 4.80 \times 10^{-16}$ J
Speed: $v = \sqrt{\frac{2E_k}{m_e}} = \sqrt{\frac{2 \times 4.80 \times 10^{-16}}{9.11 \times 10^{-31}}} = 3.25 \times 10^{7}$ m s^{-1}
de Broglie wavelength:

$$\lambda = \frac{h}{m_e v} = \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \times 3.25 \times 10^{7}} = \mathbf{2.24 \times 10^{-11}} \text{ m}$$

This is comparable to atomic spacings ($\sim 10^{-10}$ m), confirming why electron diffraction is observed.

Example 3 — Energy Level Transition

Question: An electron in a hydrogen-like atom falls from an energy level of -1.5 eV to -3.4 eV. Find the frequency and wavelength of the emitted photon.

Solution

Solution:
Energy of emitted photon:

$$hf = E_2 - E_1 = -1.5 - (-3.4) = 1.9 \text{ eV} = 1.9 \times 1.60 \times 10^{-19} = 3.04 \times 10^{-19} \text{ J}$$

Frequency:

$$f = \frac{3.04 \times 10^{-19}}{6.63 \times 10^{-34}} = \mathbf{4.59 \times 10^{14}} \text{ Hz} \quad \text{(visible light — red/orange)}$$

Wavelength:

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^{8}}{4.59 \times 10^{14}} = \mathbf{654} \text{ nm}$$

Practice Exam Questions

Section A — Short Answer Questions

Q1. State what is meant by the *threshold frequency* in the photoelectric effect.

[2 marks]

Q2. Explain why the photoelectric effect cannot be explained by the wave model of light. Give two reasons.

[2 marks]

Q3. Light of frequency 8.0×10^{14} Hz is incident on a metal surface with work function 2.5 eV. Show whether photoelectric emission will occur and, if so, calculate the maximum kinetic energy of the emitted electrons.

[4 marks]

Q4. Calculate the de Broglie wavelength of a proton ($m_p = 1.67 \times 10^{-27}$ kg) moving at 2.0×10^6 m s^{-1}.

[2 marks]

Q5. An X-ray tube operates at an accelerating voltage of 80 kV. Calculate the minimum wavelength of X-rays produced.

[3 marks]

Section B — Longer Structured Questions

Q6. A student investigates the photoelectric effect using a metal surface. The graph of maximum kinetic energy against frequency is a straight line. The threshold frequency of the metal is 5.5×10^{14} Hz.

(a) Calculate the work function of the metal in joules.

[2 marks]

(b) Light of frequency 9.0×10^{14} Hz is incident on the surface. Calculate the stopping potential.

[3 marks]

(c) The intensity of the incident light is doubled whilst keeping its frequency the same. State and explain what happens to (i) the rate of electron emission, and (ii) the maximum kinetic energy of emitted electrons.

[4 marks]

Q7. The energy levels of atomic hydrogen include $E_1 = -13.6$ eV (ground state), $E_2 = -3.4$ eV, and $E_3 = -1.5$ eV.

(a) Calculate the frequency of the photon emitted when an electron transitions from $n = 3$ to $n = 1$.

[3 marks]

(b) A photon of energy 12.1 eV is incident on a hydrogen atom in the ground state. Explain whether this photon will be absorbed.

[2 marks]

(c) Describe how the line spectrum of hydrogen provides evidence for the existence of discrete energy levels in atoms.

[3 marks]

Mark Scheme and Answers

Q1. The threshold frequency is the minimum frequency of electromagnetic radiation [1] that is capable of causing photoelectric emission from the surface [1].

Q2. Any two from: wave theory predicts a time delay before emission, but emission is instantaneous [1]; wave theory predicts any frequency of sufficient intensity should cause emission, but no emission occurs below f_0 regardless of intensity [1]; wave theory predicts intensity should increase electron KE, but maximum KE depends only on frequency [1].

Q3. Photon energy: $E = hf = 6.63 \times 10^{-34} \times 8.0 \times 10^{14} = 5.30 \times 10^{-19}$ J $= 3.32$ eV [1]. Since 3.32 eV > 2.5 eV, emission **will** occur [1]. Maximum KE $= 3.32 - 2.5 = 0.82$ eV $= 1.31 \times 10^{-19}$ J [2].

Q4. $\lambda = h/(m_p v) = 6.63 \times 10^{-34}/(1.67 \times 10^{-27} \times 2.0 \times 10^6) = \mathbf{1.98 \times 10^{-13}}$ m [2].

Q5. $\lambda_{\text{min}} = hc/eV = (6.63 \times 10^{-34} \times 3.00 \times 10^8)/(1.60 \times 10^{-19} \times 80 \times 10^3)$ [2] $= \mathbf{1.55 \times 10^{-11}}$ m [1].

Q6(a). $\phi = hf_0 = 6.63 \times 10^{-34} \times 5.5 \times 10^{14} = \mathbf{3.65 \times 10^{-19}}$ J [2].

Q6(b). $E_{k(\text{max})} = hf - \phi = 6.63 \times 10^{-34} \times 9.0 \times 10^{14} - 3.65 \times 10^{-19} = 5.97 \times 10^{-19} - 3.65 \times 10^{-19} = 2.32 \times 10^{-19}$ J [2]; $V_s = E_{k(\text{max})}/e = 2.32 \times 10^{-19}/1.60 \times 10^{-19} = \mathbf{1.45}$ V [1].

Q6(c). (i) Rate of electron emission **doubles** [1]: greater intensity ⇒ more photons per second ⇒ more photoelectron emissions per second [1]. (ii) Maximum KE is **unchanged** [1]: each photon still has the same energy (f unchanged); intensity affects number of photons, not energy per photon [1].

Q7(a). $\Delta E = -1.5 - (-13.6) = 12.1$ eV $= 12.1 \times 1.60 \times 10^{-19} = 1.936 \times 10^{-18}$ J [1]; $f = \Delta E/h = 1.936 \times 10^{-18}/6.63 \times 10^{-34} = \mathbf{2.92 \times 10^{15}}$ Hz (ultraviolet) [2].

Q7(b). The energy difference from $n = 1$ to $n = 3$ is $-1.5 - (-13.6) = 12.1$ eV [1]; the photon energy exactly matches this transition, so it **will** be absorbed and the electron promoted to $n = 3$ [1].

Q7(c). Each line in the emission spectrum corresponds to a specific photon frequency [1]; each frequency corresponds to a specific photon energy, given by $hf = E_2 - E_1$ [1];

the existence of discrete lines (rather than a continuum) shows that electrons can only transition between fixed energy values — i.e. energy levels are quantised [1].

Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

	Learning Objective	Confidence (1–3)
☐	State that electromagnetic radiation is quantised into photons	
☐	Use $E = hf$ and $E = hc/\lambda$ to calculate photon energy	
☐	Convert between joules and electronvolts	
☐	Describe the photoelectric effect and state its key observations	
☐	Explain why wave theory fails to explain the photoelectric effect	
☐	Apply Einstein's equation $hf = \phi + \frac{1}{2}mv_{\text{max}}^2$	
☐	Define work function and threshold frequency; relate via $\phi = hf_0$	
☐	Explain the significance of stopping potential and use $eV_s = E_{k(\text{max})}$	
☐	State and apply the de Broglie relation $\lambda = h/p$	
☐	Describe electron diffraction as evidence for wave–particle duality	
☐	Explain what is meant by atomic energy levels and ground/excited states	
☐	Use $hf = E_2 - E_1$ for emission and absorption transitions	
☐	Interpret emission and absorption line spectra	
☐	Explain how line spectra provide evidence for energy quantisation	
☐	Describe the origin of continuous and characteristic X-ray spectra	
☐	Use $\lambda_{\text{min}} = hc/eV$ for minimum X-ray wavelength	

Key: 1 = Need more work 2 = Getting there 3 = Confident

Good luck with your revision!

Quantum physics is genuinely strange — but the maths is manageable. Focus on understanding *why* classical physics fails, and let the photon model guide the rest. The equations follow naturally from the physics.